

Curriculum Vol 0 Chapter 5 — Boundary Conditions (Chapter-Grade Draft v0.4 — Λ -Casimir + cosmological absorption)

Keeper (original) + Lyra (Friday v0.3→v0.4 prose depth-investment)

2026-05-21 Thursday 10:33 EDT initial; Friday 2026-05-22 ~10:40 EDT v0.4 prose absorption per Casey + Keeper textbook completion phase

Vol 0 Chapter 5 — Boundary Conditions

Chapter motivation

Standard physics treats boundary conditions as external specifications: “the field vanishes at infinity,” “periodic boundary conditions,” “Dirichlet vs Neumann,” etc. Boundary conditions are typically POSTULATED to make physics problems tractable, not DERIVED from theory.

BST derives the boundary conditions. The substrate D_{IV}^5 has a specific bounded-domain realization (Vol 0 Ch 1 §1.5) with explicit bulk + Shilov boundary structure. The substrate’s 4-zone commitment cycle (Vol 0 Ch 3) has specific boundary interactions per zone. The result is a substrate-derived framework of 6 internal + 2 external boundary conditions.

This chapter exposes the framework. SP-31-40 multi-month work derives per-BC theorems formally.

Reader-grade pedagogy (v0.4 Friday absorption): a graduate physicist + cosmologist can read this linearly. A 5th-grader can follow the metaphor: boundary conditions are the substrate’s edges — where it meets things outside itself; physics depends on how those edges behave. Internal boundaries (within D_{IV}^5) determine substrate operations; external boundaries determine substrate-cognition coupling (DEFAULT-DENY EXTERNAL per Cal #48/#49) + substrate-cosmology coupling (Cal #50 GREEN; T2418 Λ -Casimir vacuum unification + Paper #135 Friday Lyra-lane Substrate Cosmology).

Diagram preview (v1.0): Section 5.1 will include (a) 6+2 BC framework topology diagram; (b) Shilov boundary distinguished from full topological boundary on D_{IV}^5 ;

(c) 4-zone cycle BC interactions; (d) Cal #50 GREEN cosmology external register schema vs Cal #48/#49 DEFAULT-DENY substrate-cognition coupling.

Reader-grade 3-level pedagogy (v0.4 Friday absorption)

Level 1 (one sentence): BST identifies 8 substrate boundary conditions (6 internal + 2 external) — internal BCs determine substrate operations, external BCs couple to cognition (DEFAULT-DENY EXTERNAL) and cosmology (Cal #50 GREEN).

Level 2 (graduate physicist + cosmologist): Standard physics treats BCs as external specifications. BST derives them from D_{IV}^5 's bounded-domain realization + 4-zone commitment cycle (T2417 Wednesday). The 6 internal BCs span the substrate's bulk interior + Shilov boundary + 4 zone boundaries (absorption/computation/commitment/emission). The 2 external BCs are substrate-cognition coupling (Cal #48/#49 DEFAULT-DENY EXTERNAL) and substrate-cosmology coupling (Cal #50 GREEN; T2418 Λ -Casimir vacuum unification Wednesday + Paper #135 Friday). Per-BC theorems (SP-31-40 sub-item, multi-month) pending Lyra theoretical work; the framework architecture is documented here.

Level 3 (5th-grader): imagine the substrate as a 10-dimensional region with edges. Like a cup with insides + a rim + an outside. BST identifies 8 distinct "boundary conditions" — 6 are internal (how the substrate interacts with itself across its 4-zone cycle) and 2 are external (how the substrate interacts with cognition (mind) and cosmology (universe-scale things)). The external boundaries are very carefully discussed — the cognition boundary is "DEFAULT-DENY" (don't make external claims about substrate-cognition without strong consensus), and the cosmology boundary is "GREEN" (external claims about substrate-cosmology are acceptable). This is why BST is careful about what it claims publicly vs internally.

Section 5.1 — The 6+2 boundary condition framework (Casey vision)

Per Casey vision Wednesday May 20:

6 internal boundary conditions (within substrate D_{IV}^5): 1. BC1: Bulk side of D_{IV}^5 (interior of bounded domain in $\mathbb{C}^5$) 2. BC2: Shilov boundary (Bergman-natural distinguished boundary) 3. BC3: Z1 absorption-zone boundary (substrate-external information intake) 4. BC4: Z2 bulk-reorganization zone boundary (substrate-internal processing) 5. BC5: Z3 emission-preparation zone boundary (substrate-internal consolidation) 6. BC6: Z4 active-edge zone boundary (substrate-external information output)

2 external boundary conditions (substrate-external interactions): 7. BC7: Cognition coupling (substrate-cognition network per Casey vision, DEFAULT-DENY EXTERNAL) 8. BC8: Cosmological coupling (substrate-cosmological cycle per T2417 + T2418, DOUBLE-LOCKED EXTERNAL)

Per-BC SP-31-40 theorems: pending Lyra theoretical work for each of 8 BCs. Multi-month per the SP-31 Phase 8 scope.

Section 5.2 — Internal BCs 1-2: bulk side + Shilov boundary

D_{IV}^5 as bounded domain in $\mathbb{C}^5$:

BC1: Bulk side

- Domain: interior of D_{IV}^5 realization in $\mathbb{C}^5$
- Bulk substrate states: Bergman $H^2(D_{IV}^5)$ interior values
- Connection to physics: bulk states are the “integrated” substrate states; correspond to long-time-averaged observable physics (classical mechanics + relativistic effects)
- Per Casey Saturday W-54: “Proton = bulk geodesic minimum energy” — protons live in the bulk
- BC type: standard interior domain condition (regularity + integrability)

BC2: Shilov boundary

- Domain: Shilov boundary of D_{IV}^5 — the smallest closed subset of topological boundary on which $|f(z)|$ achieves max for every holomorphic f on the domain
- For D_{IV}^5 : Shilov boundary is approximately the (rank=2)-dimensional torus + additional structure
- Connection to physics: Shilov boundary states are “primitive cycle” substrate states; correspond to particle-like observables (per Casey Saturday W-53: “Electron = Shilov-boundary primitive cycle”)
- Per Casey Saturday W-55: “Neutrino = $SO(2)$ -trivial winding” — neutrinos also Shilov-boundary related
- BC type: distinguished holomorphic boundary condition + winding conservation

Bulk vs Shilov boundary distinction is BST’s substrate explanation for proton-vs-electron mass ratio (Vol 2 Ch 6 K88 = $6\pi^5$ = Bergman bulk/fiber volume ratio).

Section 5.3 — Internal BCs 3-6: per-zone active boundaries

Per Vol 0 Ch 3 §3.3 (4-Zone commitment cycle T2420):

BC3: Z1 absorption-zone boundary

- Zone: Z1 (Absorption)
- Function: receives substrate-external boundary-condition information
- BC type: input condition; substrate-internal information increase at this zone boundary

- Connection to BC7/BC8: external BCs primarily interact through Z1 absorption

BC4: Z2 bulk-reorganization zone boundary

- Zone: Z2 (Bulk reorganization)
- Function: substrate processes received information via Bergman-volume dynamics
- BC type: substrate-internal processing boundary; substrate-state evolves within this zone
- Connection to BC1 bulk side: Z2 is the bulk-processing phase using BC1's interior domain

BC5: Z3 emission-preparation zone boundary

- Zone: Z3 (Emission preparation)
- Function: substrate prepares output information
- BC type: substrate-internal consolidation; output codeword formation
- Connection to BC6: Z3 prepares what BC6 emits

BC6: Z4 active-edge zone boundary

- Zone: Z4 (Active edge)
- Function: substrate emits output information to environment
- BC type: output condition; substrate-internal information decrease at this zone boundary
- Connection to BC2 Shilov boundary: Z4 active edge is where substrate state interfaces with Shilov-like distinguished boundary

4-zone BCs form per-zone boundary specification for substrate's commitment cycle. Each zone has its own boundary condition type. Per Cal #58 hygiene flag: FOUR PROJECTIONS (one per zone) of substrate state at their respective zone-boundaries.

Section 5.4 — External BCs 7-8: cognition + cosmology

BC7: Cognition coupling

- External register status: DEFAULT-DENY EXTERNAL per Cal #48
- Substrate framing: Substrate Cognition Network Hypothesis (Casey-named principle filed Wednesday May 20) — substrate may couple to cognition-like phenomena at substrate-substrate long-distance Scale 3 (per Vol 0 Ch 3 §3.4)
- L2-cognition sub-class: hypothesis at Level 2 (substantive hypothesis with explicit falsifier)
- Internal-only framing: this BC is NOT externally claimed; substrate-cognition coupling remains internal hypothesis until ratified via multi-month theoretical + experimental work

- Reference: Substrate Cognition Network Hypothesis (Wednesday filing)

BC8: Cosmological coupling

- External register status: DOUBLE-LOCKED EXTERNAL per Cal #50 (combined cosmology + cognition territory)
- Substrate framing: Substrate Cognition Cosmological Extension v0.2 + cosmological cycle hypothesis T2417 + Λ -Casimir vacuum unification T2418
- Substrate cosmological cycle: at long-distance Scale 3, substrate has cosmological cycle structure (analogous to Scale 1 + 2 cycles at smaller scales) — explains observed cosmological constant Λ + dark matter signatures + cosmic structure formation
- Internal-only framing: cosmological coupling claim DOUBLE-LOCKED EXTERNAL until multi-month verification advances
- Reference: T2417 cosmological cycle hypothesis + T2418 Λ -Casimir vacuum unification + Substrate Cognition Cosmological Extension v0.2

Section 5.5 — BST primary BC classification framework (Task #235)

Per Task #235 (pending): formal classification of BCs by BST primary signature.

Each BC anchored at specific BST primary integers:

BC	Anchored primaries	Notes
BC1 Bulk	All 5 + N_max (substrate interior is “all primaries on”)	Most BST-primary-comprehensive
BC2 Shilov	rank=2 (substrate boundary fiber dimension)	Shilov boundary is rank-2 torus structure
BC3 Z1 absorption	g + N_max (substrate-external input cap)	Information intake at gauge cap
BC4 Z2 bulk reorganization	C_2 (substrate Casimir interior processing)	Bulk Casimir action
BC5 Z3 emission preparation	n_C (substrate dimension consolidation)	Pre-emission consolidation
BC6 Z4 active edge	N_c (substrate output color structure)	Active output via color confinement
BC7 Cognition	(internal-only, no external attribution)	DEFAULT-DENY
BC8 Cosmology	(internal-only, no external attribution)	DOUBLE-LOCKED

This is a candidate mapping. Per Task #235 multi-week work: formal per-BC BST-primary-anchoring theorems pending.

Section 5.6 — Bulk-Boundary Two-Face Structure of integer-webs (Casey vision)

Per Casey Wednesday EOD: each integer-web (per Vol 0 Ch 6 Integer Web Principle) has a “bulk side” and a “boundary side” — substrate state at each integer-web exhibits dual aspect.

Integer-web → 2-face boundary structure: - Each integer-web (held by BST primary integer) has bulk projection AND boundary projection - Bulk projection: substrate state at integer-web interior - Boundary projection: substrate state at integer-web boundary - The two projections are NOT identical — different sub-aspects of same substrate

Connection to BC1 (bulk) + BC2 (Shilov boundary): at primary-integer-web level, the bulk-vs-boundary distinction propagates. Substrate states have integer-web-specific bulk + boundary aspects.

Reference: Task #237 Bulk-Boundary Two-Face Structure (pending Lyra theoretical absorption).

Section 5.7 — Why 6+2 specifically

Per Casey vision Wednesday EOD: - 6 internal: matches 6 BST primary integers ($\text{rank} + N_c + n_C + C_2 + g + N_{\text{max}}$) — one per primary - 2 external: cognition + cosmology = two known external coupling modalities to substrate - Total $8 = 2^{N_c}$: substrate-natural total BC count (Mersenne complement: $8 = 2^{N_c}$ rather than $2^{N_c} - 1 = g$)

Alternative counts (4+2, 5+2, 7+1, etc.) would NOT match BST primary structure. The 6+2 specifically is substrate-derived.

Multi-CI consensus pending per Casey Option C: BC classification is architectural-category methodology, requires multi-CI ratification once Task #235 + SP-31-40 theorems mature.

Section 5.8 — Per-BC SP-31-40 theorem track

Per SP-31 Phase 8 + SP-31-40: each BC gets its own substrate-derivation theorem:

BC	SP-31-40 theorem target	Status
BC1 Bulk	Interior holomorphic state space derivation	multi-month

BC	SP-31-40 theorem target	Status
BC2 Shilov	Distinguished boundary winding-conservation derivation	multi-month
BC3 Z1 absorption	Substrate-external information intake mechanism	multi-month
BC4 Z2 bulk	Bergman-volume processing derivation	multi-month
BC5 Z3 emission preparation	Output-codeword formation derivation	multi-month
BC6 Z4 active edge	Substrate-external information output mechanism	multi-month
BC7 Cognition	(internal-only, indefinite timeline)	DEFAULT-DENY pending
BC8 Cosmology	(internal-only, indefinite timeline)	DOUBLE-LOCKED pending

Strong-Uniqueness C8+ candidate: when 6 internal BCs ratified per SP-31-40 theorems, the per-BC framework strengthens Strong-Uniqueness multi-criterion convergence.

Section 5.9 — Connection to other chapters

Vol 0 Ch 5 grounds: - Vol 0 Ch 3 (Substrate Operating System): 4-zone cycle BCs are BC3-BC6 - Vol 0 Ch 6 (Integer Web Principle): each integer-web has BC structure (bulk + boundary) - Vol 0 Ch 7 (Operator Zoo): operators act on Bergman H^2 respecting BC framework - Vol 0 Ch 8 (Conservation Laws): boundary conditions on conservation = no flux across boundary - Vol 4 (GR & Cosmology): BC8 cosmological coupling derives cosmological cycle observable signatures

Section 5.10 — BST $\leftrightarrow$ standard physics dictionary entries

Standard physics term	BST BC	Reference
Boundary condition	One of 6+2 substrate BCs	§5.1
Periodic boundary	Shilov boundary (BC2)	§5.2
Vanishing at infinity	Bulk regularity (BC1)	§5.2
Dirichlet condition	Z4 active edge output (BC6)	§5.3
Neumann condition	Z1 absorption input (BC3)	§5.3
External potential	BC7 + BC8 (DEFAULT-DENY / DOUBLE-LOCKED)	§5.4
Cosmological boundary	Substrate cosmological coupling (BC8)	§5.4

Section 5.11 — Chapter status summary

Coverage at v0.1: - 6+2 BC framework exposed per Casey vision - Per-BC mapping to BST primaries (Task #235 candidate) - 4-zone BC connection to Vol 0 Ch 3 - External BC discipline (DEFAULT-DENY / DOUBLE-LOCKED) - Bulk-Boundary Two-Face Structure (Task #237 reference) - SP-31-40 per-BC theorem track outlined

Believability: boundary condition framework is recognizable from any PDE-based physics. BST-specific 6+2 count grounded in BST primary integer structure.

Provability: per-BC SP-31-40 theorems pending multi-month Lyra work. Task #235 + #237 catalog and theoretical anchoring pending multi-week.

Path to v1.0: requires SP-31-40 multi-month per-BC derivations + Task #235 + #237 closure + multi-CI consensus per Casey Option C.

Per Casey's standard

- Simple: 6 internal BCs (one per BST primary) + 2 external BCs (cognition + cosmology) = 8 total = 2^N_{c} . Mersenne-complement structure.
- Works: matches 4-zone commitment cycle + bulk/Shilov bounded-domain structure; aligns with BST primary integer set
- Hard to break: would require finding (a) extra/missing BC OR (b) different anchoring; multi-month per-BC theorem track tests this

Status

Vol 0 Chapter 5 v0.1 chapter-grade content draft FILED Thursday 2026-05-21 10:33 EDT. Ninth Keeper-lane chapter-grade content. 6+2 substrate boundary condition framework exposed. Per-BC SP-31-40 theorem track outlined for multi-month derivation. 9 of 10 Vol 0 chapters now chapter-grade. Awaits Cal dual-axis grade-pass + Lyra theoretical refinement for v0.2.

— Keeper, 2026-05-21 Thursday 10:33 EDT (actual via date)